MILITARY SPECIFICATION

CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM, OR PLASTIC FILM DIELECTRIC, DIRECT AND ALTERNATING CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-39022D, dated 12 June 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Following title, insert: "FAILURE RATE LEVEL 'L' IS INACTIVE FOR NEW DESIGN AFTER 20 July 1983."

1.1, lines 6 and 7: Following "producer's risk and", add "unless otherwise specified (see 3.1)"; and add "Unless otherwise specified (see 3.1)," at beginning of last sentence, and "(See 6.4.)" at end of last sentence.

PAGE 2

2.1, list of Military Standards, add:

"MIL-STD-1276 - Leads For Electronic Component Parts."

PAGE 3

3.5.1.1: Delete first sentence and substitute "Axial-wire lead terminals shall be of copper, copper alloy, copper clad steel, or (when specified, see 3.1), type C or N2 of MIL-STD-1276. The size shall be as specified (see 3.1)."

3.7: Following title, add "(not applicable to CHR01 styles)." Make this same change on page 7, paragraph 3.27.

PAGE 4

3.10, line 3: Following last sentence, add "unless otherwise specified (see 3.1)." Make this same change in paragraph 3.17, line 2; paragraph 3.20, line 2 (page 5); paragraph 3.22, line 5 (page 5); paragraph 3.24, line 2 (page 6); and paragraph 3.26, line 2 (page 6).

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3.28, delete and substitute:

"3.28 Marking. Marking of capacitors shall conform to method I of MIL-STD-1285, and shall include the part number, 'JAN' marking, date code, lot symbol, trademark (CHR01 styles only), and manufacturer's source code. Except for CHR01 styles, the capacitance (in µF), capacitance tolerance, and rated voltage shall also be included. The date code shall designate the week that the items are offered for inspection during the period specified in 4.6.1.1. The lot symbol shall be a single letter except that letters 'O' and 'I' shall not be used (the letter 'A' shall be used to identify a single lot and the date code will indicate the lot). Manufacturing records shall include these same date codes and lot symbols. The manufacturer shall provide for lot traceability by date code and lot symbol. Example of marking (unless otherwise specified, see 3.1):

M39022/01-1001	-	Part number.
JAN7833A 12345	-	'JAN' marking, date code, lot symbol, and source code.
.018 µF 5% 50 V	-	Capacitance, capacitance tolerance, and rated voltage.

The date and source codes shall also be marked on the unit package."

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4.2b., delete and substitute:

"b. Verification of qualification (see 4.5)."

4.3.1, line 4: Following last sentence, add "unless otherwise specified (see 3.1)."

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4.5, delete and substitute:

"4.5 Verification of qualification. Every six months, the manufacturer shall compile a summary of the results of quality conformance inspections and, where applicable, extended failure rate (FR) test data, in the form of a verification of qualification report, and forward it to the qualifying activity as the basis of continued qualification approval. In addition, within 30 days of the end of the reporting date, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain his qualified FR level. Continuation shall be based on evidence that, over the six month period, the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so as to affect the character of the item, (not applicable to CHR01 styles).

d. Lot rejection for group A inspection does not exceed 10 percent or one lot, whichever is greater (not applicable to CHR01 styles).

e. The requirements for groups B and C inspections are met.

f. The records of all FR tests combined substantiate that the "M" (1.0 percent) or "P" (0.1 percent) FR level has been maintained, or that the manufacturer continues to meet the "R" (0.01 percent) and "S" (0.001 percent) FR level for which qualified, although the total component hours of testing does not, as yet, meet the requirements of 4.4.4c.

If group C test requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 30 days after completion of retesting of the group C inspections. In this case, the qualifying activity shall be notified of this condition within the time the original verification of qualification report was due. All reports shall be certified by a responsible company official. The qualifying activity shall be contacted for the report format.

4.5.1 Records. Maintenance of life test and FR level records shall be as specified in MIL-STD-690."

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4.6.1.2: Following last sentence, add "Prior to group A inspection, CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N styles shall be exposed to a temperature of +125°C with 140-percent of dc rated voltage for 48 hours, minimum; CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P styles shall be exposed to a temperature of +125°C with 140-percent of dc rated voltage for 16 hours, minimum."

4.6.1.2.1: Following title, add "(not applicable to CHR01 styles)."

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Add the following:

"4.6.1.2.1.1 CHR01 styles. 100-percent inspection shall apply to all FR levels for subgroups 2 and 3; however, in subgroup 3, only 5 sample units need be inspected for physical dimensions."

4.6.1.2.2, line 2: Following "subgroup 2" add "(and subgroup 3, for CHR01 styles)". Make this same change in paragraph 4.6.1.2.2a., line 2.

4.6.1.2.3: Following title, add "(not applicable to CHR01 styles)."

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TABLE IV, Subgroup 2: Following "Thermal shock", add footnote "3/"; following "Insulation resistance", add "(terminal to terminal, CHR01 styles only)"; and add the following footnote at end of table:

"3/ 10 cycles, min (CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, and CHR01P).
5 cycles, min (CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R)."

4.6.1.3: Following title, add "(not applicable to CHR01 styles)."

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4.6.2.1.1: In line 2, following "every 2 months", add "(for CHR01 styles, 18 sample units shall be taken from production every 5 months and shall consist of CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N styles only)"; and following last sentence, add "For subgroup 3, a minimum of 5 sample units for each CHR01 style shall be subjected to the tests specified, in the order shown."

* 4.7.2.a, delete and substitute:

"a. 140 percent of rated rms voltage at 400 hertz (Hz) shall be applied on voltages of ≤300 volts; 120 percent of rated rms voltage at 400 hertz (Hz) shall be applied on voltages ≥350 volts."

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TABLE VI: In subgroups 1A, 1B, and 1C (fourth column), add "or 6 (CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N styles)"; in Subgroup 2 title, add "(not applicable to CHR01 styles)"; and add the following at end of table:

Test	Requirement paragraph	Method paragraph	Number of sample units for inspection	Allowable failures
Subgroup 3 (every month)				
Subgroup 3A				
(All FR levels for CHR01 styles)				
Capacitance change with temperature	3.22	4.7.18	5 per style, minimum	
Life	3.26	4.7.22.3.1		2/
Subgroup 3B				
(CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N)				
AC life (240 hours)	3.26	4.7.22.4	3/	4/
Dissipation factor	3.13	4.7.9		
Insulation resistance	3.11	4.7.7		

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TABLE VI, following footnote 2/, add the following:

"3/ The number is at the option of the manufacturer; the total capacitance of the sample units to be tested shall equal 40 µF, unless otherwise specified (see 3.1).

4/ For reference only; data will be submitted to qualifying activity."

4.7.4a., line 1: Delete "A" and substitute "B".

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4.7.9: At beginning of last sentence, add "Unless otherwise specified, see 3.1,".

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4.7.22.1, line 2: Following last sentence, add "unless otherwise specified, see 3.1."

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Add the following:

"4.7.22.4 AC life (for group C, CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N). Capacitors shall be exposed to the voltages and frequencies as specified (see 3.1) at 100°C for 240 +48 -0 hours."

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6.4: At end of last sentence, add "However, CHR01 styles may be used for applications requiring a tight capacitance tolerance, excellent capacitance stability, very high insulation resistance, and low loss factors where the ac component of voltage is large with respect to the applied dc voltage and where momentary breakdowns (see 6.3.1) cannot be tolerated."

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20.1.2: Following last sentence, add "For environmental tests only, CHR01A will qualify CHR01B and CHR01C; CHR01D will qualify CHR01E and CHR01F; CHR01G will qualify CHR01H and CHR01J; CHR01K will qualify CHR01L and CHR01M; and CHR01N will qualify CHR01P and CHR01R."

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TABLE VII, delete CHR49 and substitute the following:

"

CHR49	AC rated voltage		
	165 V	M39022/07-1001	38
	110 V	M39022/07-1012	38
	80 V	M39022/07-1013	38
	200 V	M39022/07-1022	38
	400 V	M39022/07-1151	38
	400 V	M39022/07-1157	38
CHR01A	30 V	M39022/9-A349L	76
CHR01B	30 V	M39022/9-B349L	76
CHR01C	30 V	M39022/9-C349L	76
CHR01D	50 V	M39022/9-D331L	76
CHR01E	50 V	M39022/9-E331L	76
CHR01F	50 V	M39022/9-F331L	76
CHR01G	100 V	M39022/9-G295L	76
CHR01H	100 V	M39022/9-H295L	76
CHR01J	100 V	M39022/9-J295L	76
CHR01K	200 V	M39022/9-K283L	76
CHR01L	200 V	M39022/9-L283L	76
CHR01M	200 V	M39022/9-M283L	76
CHR01N	400 V	M39022/9-N127L	76
CHR01P	400 V	M39022/9-P127L	76
CHR01R	400 V	M39022/9-R127L	76

"

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